

Predictive Maintenance Using IoT and AI

*Seminar report submitted in partial fulfillment of the requirements for the
award of the degree of*

Bachelor of Technology in Computer Science and Engineering

By

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Declaration of Student

I hereby declare that the 6th Semester Seminar work entitled “**Predictive Maintenance Using IoT and AI**” in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering submitted to Department of Computer Science & Engineering, Parala Maharaja Engineering College, Berhampur is an authentic record of my work carried out. The matter embodied in this Seminar report has not been submitted to any University or Institution for any degree or diploma.

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Certificate of Approval

This is to certify that the 6th Semester Seminar report entitled “**Predictive Maintenance Using IoT and AI**” submitted by **V Lalmohan Rao Pattnaik (2321109122)** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering submitted to Department of Computer Science & Engineering, Parala Maharaja Engineering College, Berhampur is a Bonafide record of his original work. The matter embodied in this seminar report has not been submitted to any other University or Institution for any degree or diploma.

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Abstract

Predictive Maintenance Using IoT and AI

Predictive maintenance is revolutionizing industrial operations by leveraging IoT sensors and machine learning to monitor equipment, detect anomalies, and predict failures before they occur. This project integrates ESP8266 microcontrollers with temperature and vibration sensors to collect real-time data from industrial machinery. The collected data is transmitted using the MQTT protocol to a cloud platform for analysis.

The predictive model employs Isolation Forest for anomaly detection and Long Short-Term Memory (LSTM) networks for predicting the Remaining Useful Life (RUL) of equipment. These algorithms are selected for their ability to handle high-dimensional data and time-series trends effectively. The methodology includes data preprocessing for noise reduction, model training on historical failure data, and visualization of results on an interactive dashboard. This system provides early warnings for maintenance needs, reducing unplanned downtime and improving overall operational efficiency.

Keywords: Predictive maintenance, IoT, ESP8266, Isolation Forest, LSTM, MQTT, machine learning

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INTRODUCTION

Predictive maintenance is an advanced maintenance strategy that uses **IoT and AI-driven analytics** to detect potential equipment failures before they occur. Unlike traditional maintenance approaches—reactive (fix after failure) and preventive (scheduled maintenance)—predictive maintenance **minimizes downtime and reduces maintenance costs**.

This project integrates IoT-enabled **ESP8266 microcontrollers** with **DHT11 (temperature, humidity)** and **MPU6050 (vibration)** sensors to collect real-time data from industrial equipment. The collected data is transmitted via **MQTT** and stored in an **SQLite database** for further processing. **Machine learning algorithms (Isolation Forest & LSTM)** analyze the data to identify anomalies and predict the **Remaining Useful Life (RUL)** of machinery. The system's **real-time dashboard (Streamlit)** provides visual insights and triggers maintenance alerts, ensuring proactive equipment management.

IoT Factory Setup: Components and Functionality

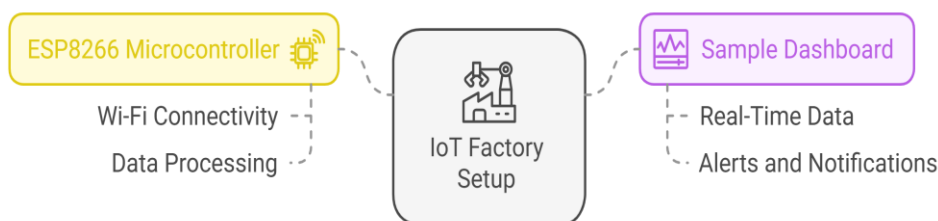


Figure 1 Basic Flow PdM Using IOT



Figure 2 IOT device & sensors

45.60 °C 59.50 % 20.38 g

🔥 High Temperature Alert! Immediate action required.

🔴 Excessive Vibration Detected! Machine might be unstable.

🔧 Next scheduled maintenance: 2025-02-25 16:14:33

Sensor Data Trends (Real-Time Historical Data)

Sensor Readings Over Time

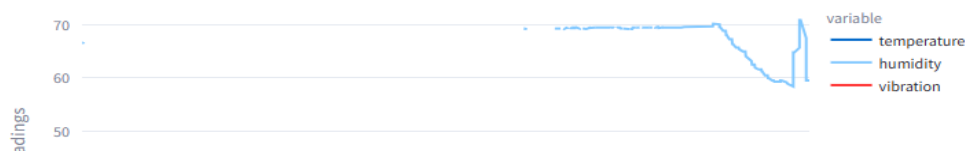
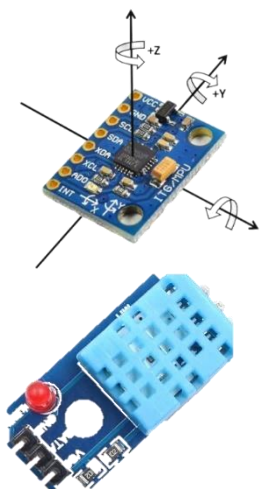


Figure 3 Dashboard



OBJECTIVE

Predictive maintenance aims to **minimize unexpected equipment failures** and **reduce maintenance costs** by using IoT and AI technologies to monitor the health of industrial machinery. Traditional maintenance methods often result in unnecessary servicing or unexpected breakdowns, leading to increased costs and downtime. This project focuses on developing an intelligent system that can **collect real-time sensor data, analyze patterns, and predict potential failures** before they occur.

Key Objectives:

- Develop an IoT-based Predictive Maintenance System to enable real-time equipment health monitoring.
 - Collect and analyze sensor data (temperature, humidity, vibration) using ESP8266, DHT11, and MPU6050.
 - Implement MQTT protocol for efficient and secure data transmission to a cloud-based storage system (SQLite).
 - Use machine learning models for predictive analysis:
 - Isolation Forest: To detect anomalies in sensor readings.
 - LSTM (Long Short-Term Memory): To predict the Remaining Useful Life (RUL) of equipment.
 - Provide a real-time monitoring dashboard using Streamlit for visualization of machine status and alerts.
 - Trigger automated alerts in case of anomaly detection or if RUL falls below a critical threshold.
 - Improve operational efficiency by reducing unplanned downtime and optimizing maintenance schedules.
- Implement an IoT-based Predictive Maintenance System for industrial equipment.

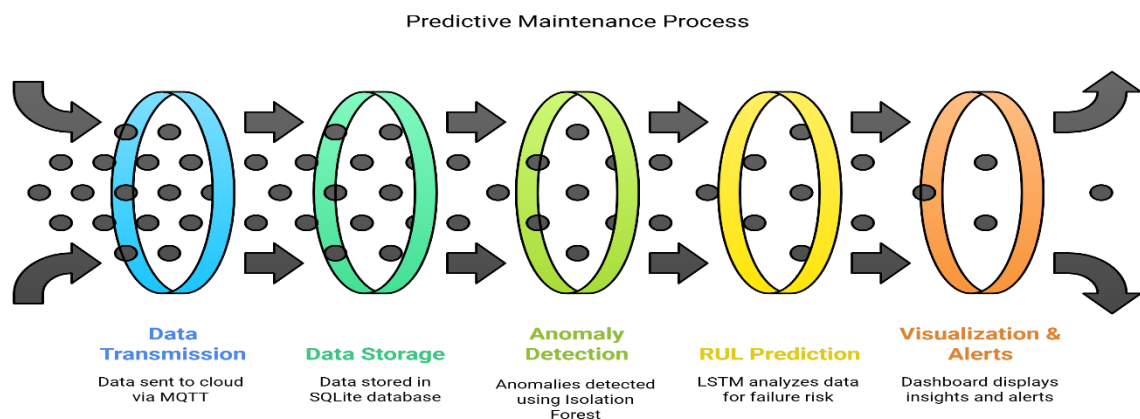


Figure 4 Process Flow

Problem Statement

Challenges in Industrial Maintenance

Industries face **frequent and unexpected equipment failures**, leading to **unplanned downtime, increased maintenance costs, and production losses**. Traditional maintenance strategies have significant drawbacks:

- **Reactive Maintenance:** Equipment is repaired only after a failure occurs, leading to costly downtime and potential safety risks.
- **Preventive Maintenance:** Scheduled maintenance is performed at regular intervals, but it does not consider real-time equipment conditions, resulting in **unnecessary servicing and resource wastage**.

Need for Predictive Maintenance

To overcome these challenges, industries require a smart, data-driven predictive maintenance system that can:

- Continuously monitor real-time sensor data from industrial machines.
- Detect anomalies in machine behaviour before failures occur.
- Predict the Remaining Useful Life (RUL) of equipment using AI models.
- Trigger maintenance alerts to prevent sudden breakdowns.

This project integrates **IoT, machine learning, and real-time data analytics** to develop a **predictive maintenance solution**, ensuring optimized maintenance scheduling, improved operational efficiency, and cost savings.

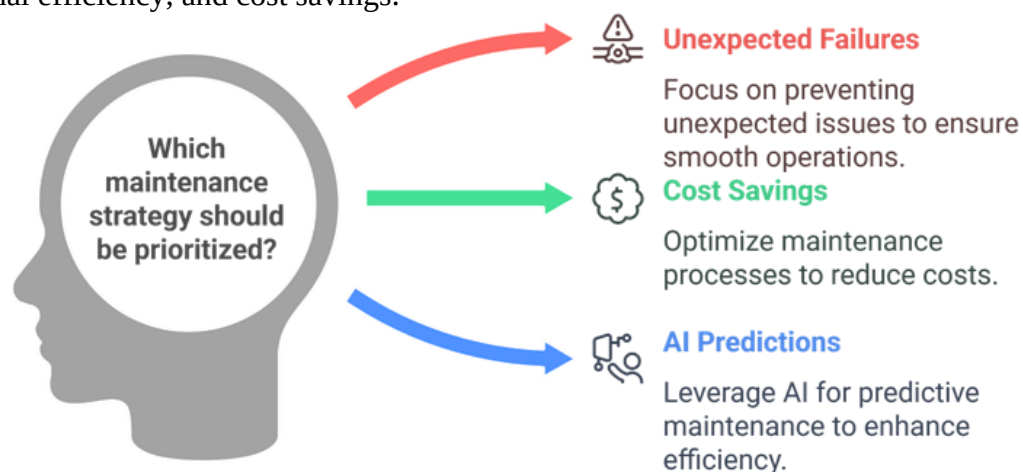


Figure 5 Problem understanding

Methodology

The Predictive Maintenance System follows a structured approach integrating **IoT, AI, and real-time analytics** to detect anomalies and predict failures in industrial equipment. The methodology consists of the following key stages:

1. Data Collection (IoT Sensors & Microcontrollers)

- **ESP8266 Microcontroller** collects real-time data from sensors.
- **DHT11 Sensor** records temperature and humidity.
- **MPU6050 Sensor** measures vibration data.

2. Data Transmission (MQTT Protocol)

- The ESP8266 sends collected sensor data to an **MQTT broker**.
- MQTT ensures **secure and efficient real-time communication**.

3. Data Storage (SQLite Database)

- A Python script (**mqtt2sqlite.py**) subscribes to the MQTT topic and stores incoming sensor data into an **SQLite database**.
- This structured storage enables historical analysis for machine learning models.

4. Anomaly Detection (Isolation Forest Model)

- The **Isolation Forest Algorithm** detects anomalies in real-time sensor readings.
- If an anomaly is detected, an **alert is triggered**.

5. Remaining Useful Life Prediction (LSTM Model)

- The **LSTM (Long Short-Term Memory) Model** analyzes time-series data.
- It predicts the **Remaining Useful Life (RUL)** of equipment based on sensor trends.

6. Visualization & Alerts (Streamlit Dashboard)

- A **Streamlit-based dashboard** provides real-time data visualization.
- **Automated alerts** notify users of anomalies or low RUL, ensuring proactive maintenance.

Working Architecture

The Predictive Maintenance System follows a structured workflow that integrates IoT, cloud communication, data storage, AI processing, and visualization to provide real-time monitoring and predictive failure detection.

1. Sensor Data Collection & Transmission

- ESP8266 reads real-time sensor data from DHT11 (Temperature, Humidity) & MPU6050 (Vibration).
- Data is sent via MQTT protocol to the cloud broker (HiveMQ).

2. Data Storage & Processing (SQLite Database)

- MQTT broker forwards sensor data to a Python script (mqtt2sqlite.py).
- Sensor readings are stored in an SQLite database for analysis.

3. Anomaly Detection & RUL Prediction

- Isolation Forest Model detects anomalies in real-time.
- LSTM Model predicts Remaining Useful Life (RUL) using past sensor trends.

4. Dashboard & Alerts (Streamlit)

- Live visualization of real-time and historical sensor data.
- Triggers alerts when an anomaly or low RUL is detected.

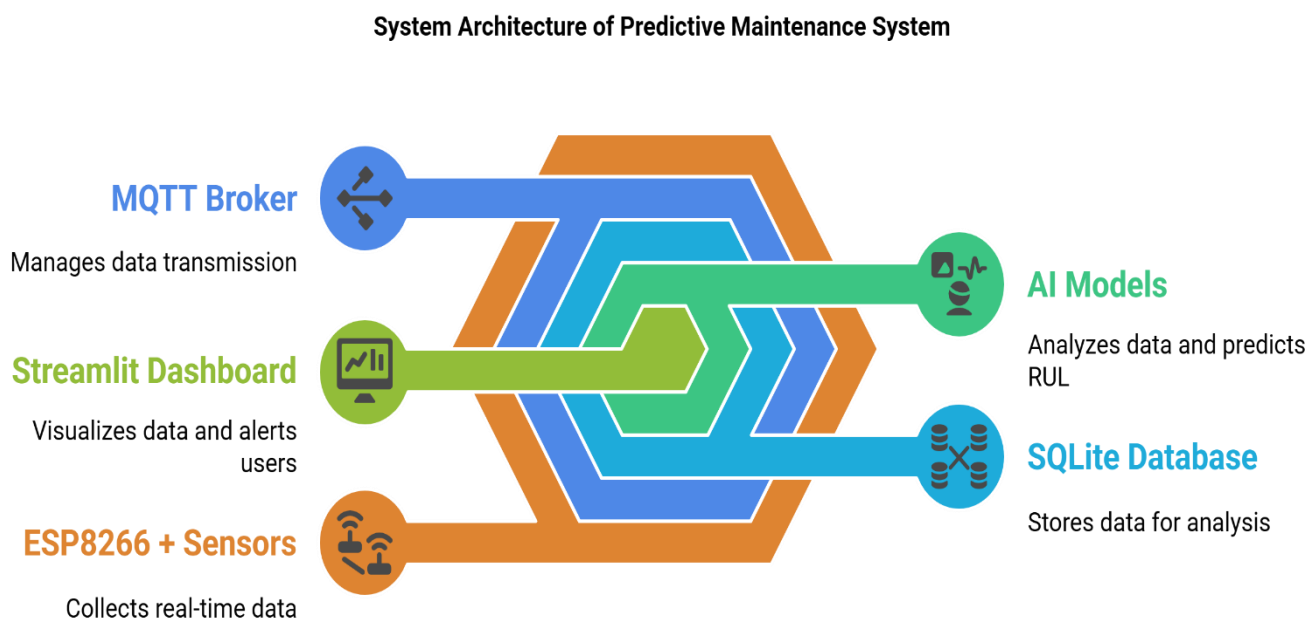


Figure 6 System Architecture

Algorithm (Pseudo Code)

The Predictive Maintenance System uses **two key machine learning algorithms** to detect anomalies and predict equipment failures:

1. Anomaly Detection - Isolation Forest Algorithm

The **Isolation Forest** model is used to identify abnormal sensor readings that may indicate a malfunction.

Input: Real-time Sensor Data (Temperature, Vibration, Humidity)

Output: Anomaly Classification (-1: Anomaly, 1: Normal)

1. Fetch latest sensor data from SQLite database.
2. Preprocess data (normalize, handle missing values).
3. Load pre-trained Isolation Forest model.
4. Predict anomaly score for incoming data:
 - If anomaly score $>$ threshold $\rightarrow$ classify as anomaly (-1).
 - Else $\rightarrow$ classify as normal (1).
5. Store anomaly classification result in SQLite database.
6. Trigger alerts if anomaly detected.

2. Remaining Useful Life (RUL) Prediction - LSTM Algorithm

The **Long Short-Term Memory (LSTM) model** is used to analyze time-series sensor data and predict the remaining lifespan of equipment

Input: Time-series Sensor Data (Temperature, Vibration, Humidity)

Output: Predicted Remaining Useful Life (RUL in days)

1. Fetch latest time-series data from SQLite database.
2. Preprocess and reshape data for LSTM input.
3. Load pre-trained LSTM model.
4. Predict RUL based on recent sensor trends.
5. Store predicted RUL values in database.
6. Display RUL predictions on dashboard.
7. Trigger maintenance alerts if $RUL <$ threshold.

Implementation

Hardware Implementation

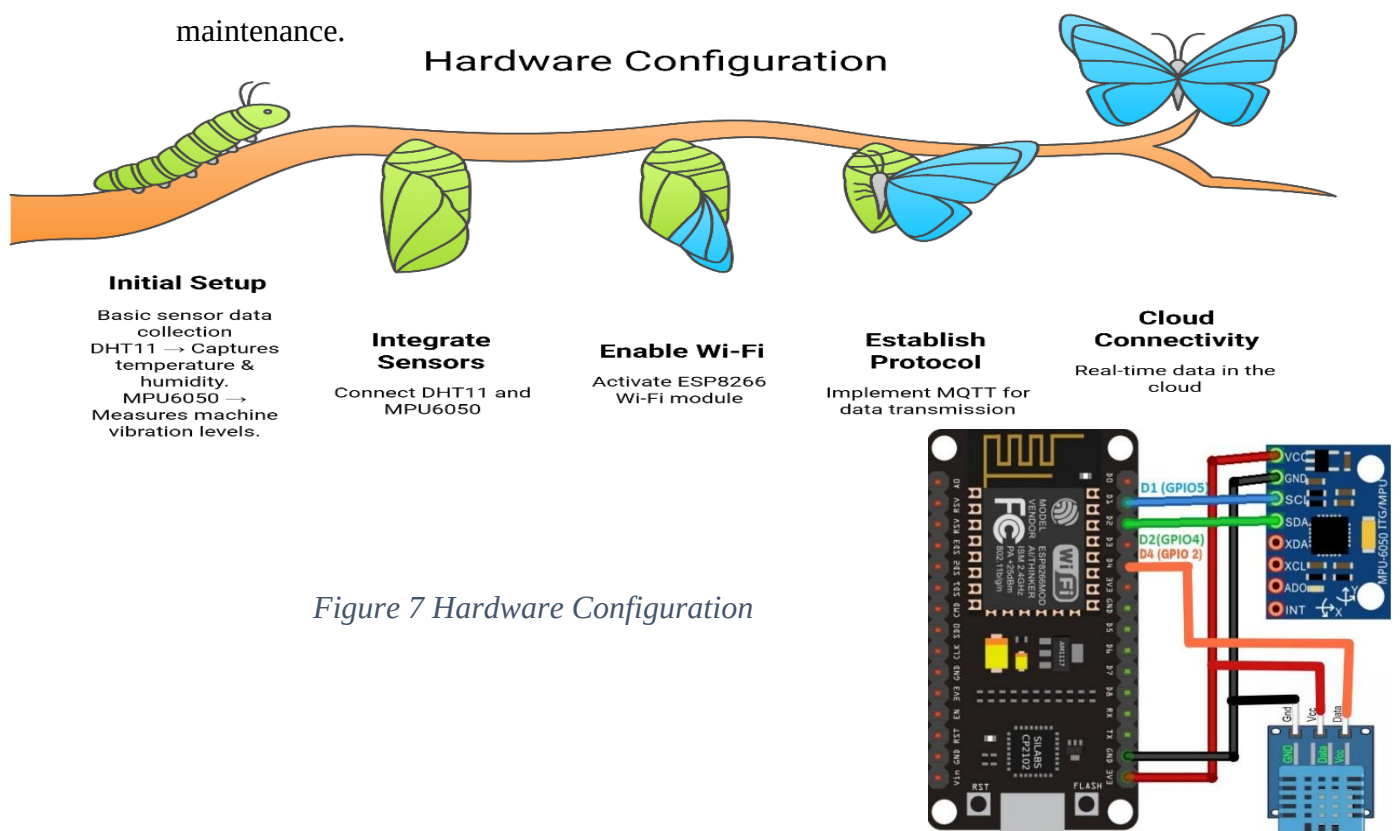
The Predictive Maintenance System uses **ESP8266** as the central microcontroller to collect and transmit real-time sensor data.

Components Used:

- **ESP8266 NodeMCU** → Microcontroller for WiFi-based communication
- **DHT11 Sensor** → Measures temperature and humidity
- **MPU6050 Sensor** → Measures vibration and motion data
- **Power Supply** → Provides stable voltage to ESP8266 and sensors

Working of Hardware:

1. **ESP8266 collects data** from DHT11 (temperature, humidity) and MPU6050 (vibration).
2. **Transmits sensor data** via MQTT protocol to the cloud broker (HiveMQ).
3. **Ensures real-time data processing** for anomaly detection and predictive maintenance.



Software Implementation

The software stack is responsible for **data processing, storage, and AI-based prediction.**

Software Components:

- **Arduino IDE** → Used to program ESP8266 for sensor data collection
- **Python & MQTT** → Handles real-time data transmission (mqtt2sqlite.py)
- **SQLite Database** → Stores timestamped sensor readings for AI analysis
- **Machine Learning Models** →
 - **Isolation Forest** → Detects real-time anomalies
 - **LSTM Model** → Predicts Remaining Useful Life (RUL)
- **Streamlit Dashboard** → Visualizes real-time sensor data and alerts

Working of Software:

1. ESP8266 transmits sensor data via MQTT.
2. Python script (mqtt2sqlite.py) captures the data and stores it in SQLite.
3. AI models analyze the data:
 - Isolation Forest detects anomalies in sensor readings.
 - LSTM predicts RUL for failure prevention.
4. Streamlit Dashboard displays alerts and predictions.

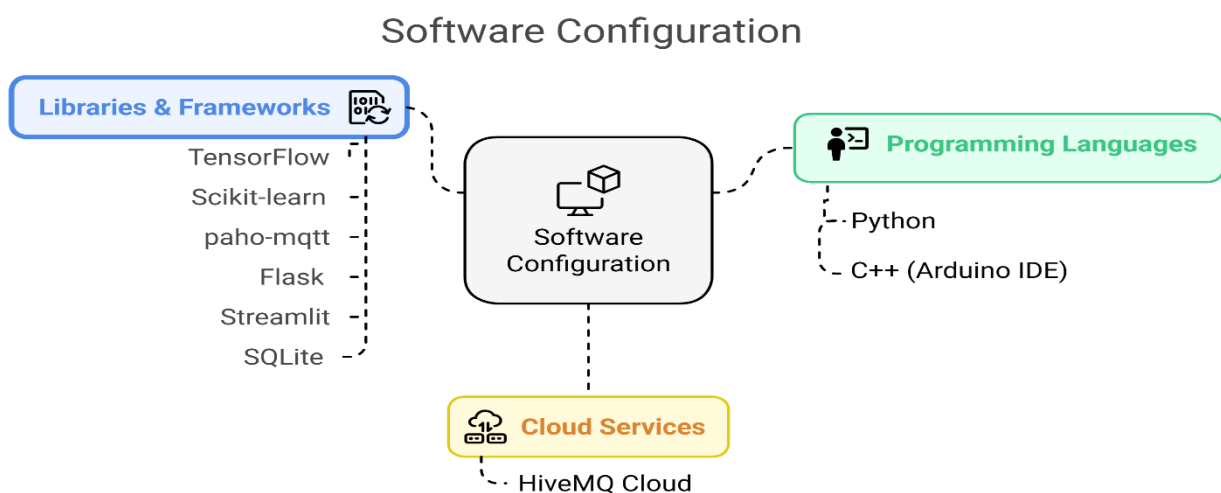


Figure 8 Software Configuration

Performance Metrics

To evaluate the system's effectiveness, we use standard Machine Learning (ML) performance metrics. These metrics help assess how well the AI models (Isolation Forest & LSTM) detect anomalies and predict failures.

1. Accuracy

- ◆ Definition: Measures how often the model's predictions are correct.
- ◆ Implementation:
 - AI models process sensor data and compare predictions with actual failure events.
 - $Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$

2. Precision

- ◆ Definition: Indicates how many of the detected anomalies are actually real.
- ◆ Implementation:
 - Isolation Forest detects anomalies in sensor readings. If an alert is raised but the equipment is fine, it's a false alarm. Higher precision means fewer false alarms.
 - Formula: $Precision = \frac{True\ Positives}{True\ Positives + False\ Positive}$

3. Recall

- ◆ Definition: Measures how many actual failures were correctly detected.
- ◆ Implementation:
 - The model checks if all true anomalies were detected.
 - If a failure occurs but the model misses it, recall is low
 - Higher recall means fewer missed failures.
 - Formula: $Recall = \frac{TruePositives}{True\ Positives + False\ Negatives}$

4. F1-Score

- ◆ Definition: Balances Precision & Recall.
- ◆ Implementation:
 - Helps in trade-off between detecting all failures (recall) and avoiding false alerts (precision).
 - Formula: $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$
 - A high F1-score ensures well-balanced anomaly detection.

5. Execution Time

- ◆ Definition: Measures how fast the system detects anomalies and updates the dashboard.
- ◆ Implementation:
 - Time is calculated from ESP8266 → MQTT → SQLite → AI Model → Dashboard update.
 - Optimizations like batch processing and efficient database queries improve speed.
 - Lower execution time ensures real-time anomaly detection.

Key Takeaways:

- ✓ Higher Accuracy → More reliable predictions.
 - ✓ Higher Precision → Fewer false alarms.
 - ✓ Higher Recall → Fewer missed failures.
 - ✓ Balanced F1-Score → Best trade-off between detection and false positives.
 - ✓ Lower Execution Time → Faster real-time response.
- ✦ Note: These metrics are implemented by evaluating AI models on historical & real-time sensor data.

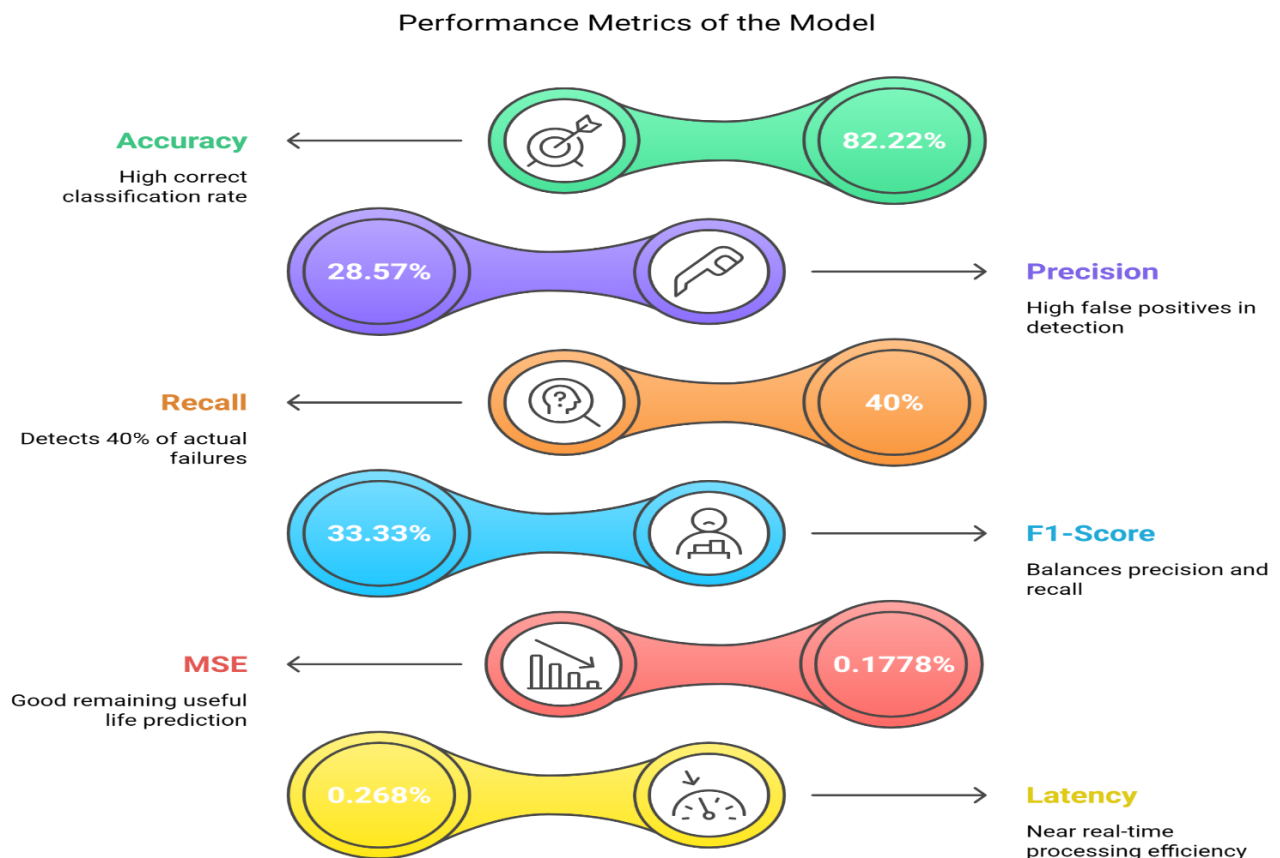


Figure 9 Performance metrics of the model

Results & Observations

The results of the **Predictive Maintenance System** are derived from real-time sensor data analysis using **AI models (Isolation Forest & LSTM)**. The system successfully detects anomalies and predicts equipment failures before they occur, reducing downtime and improving operational efficiency.

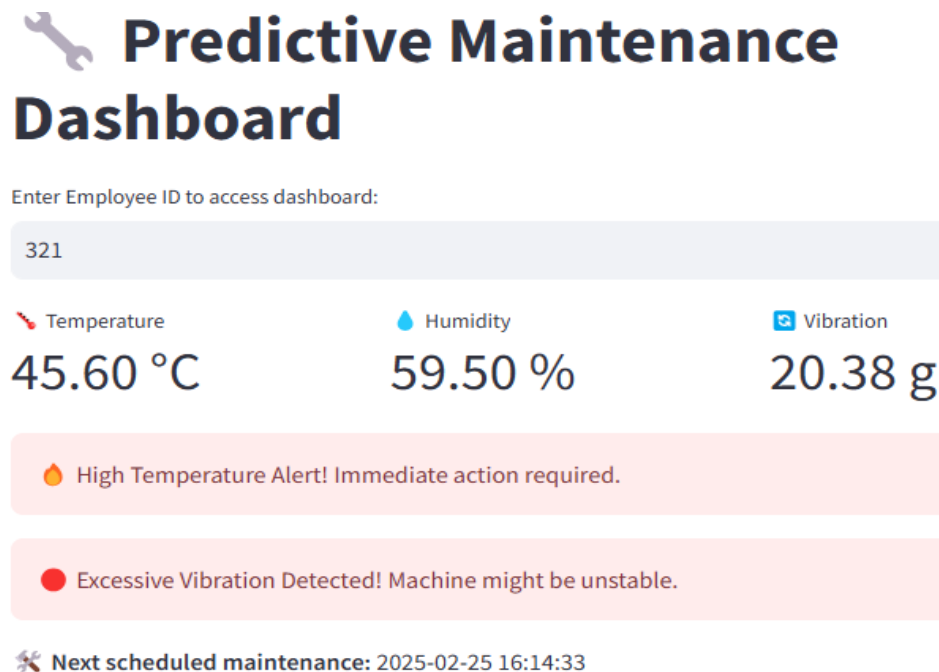


Figure 10 Result output 1

1. Predictive Maintenance Alerts

◆ Live Sensor Readings Displayed on Dashboard:

- 🌡️ **Temperature:** 45.60°C (High Alert 🚨)
- 💧 **Humidity:** 59.50%
- 📉 **Vibration:** 20.38g (Excessive Alert 🚨)

◆ Critical Alerts:

- 🔥 **High Temperature Alert:** Immediate action required.
- ⚠️ **Excessive Vibration Detected:** Machine might be unstable.
- ⌚ **Next Scheduled Maintenance:** 2025-02-25 16:14:33

📌 Observation:

- **High temperature and excessive vibration** trigger real-time alerts.
- Predictive maintenance scheduling **reduces unplanned downtime** and ensures continuous operations.

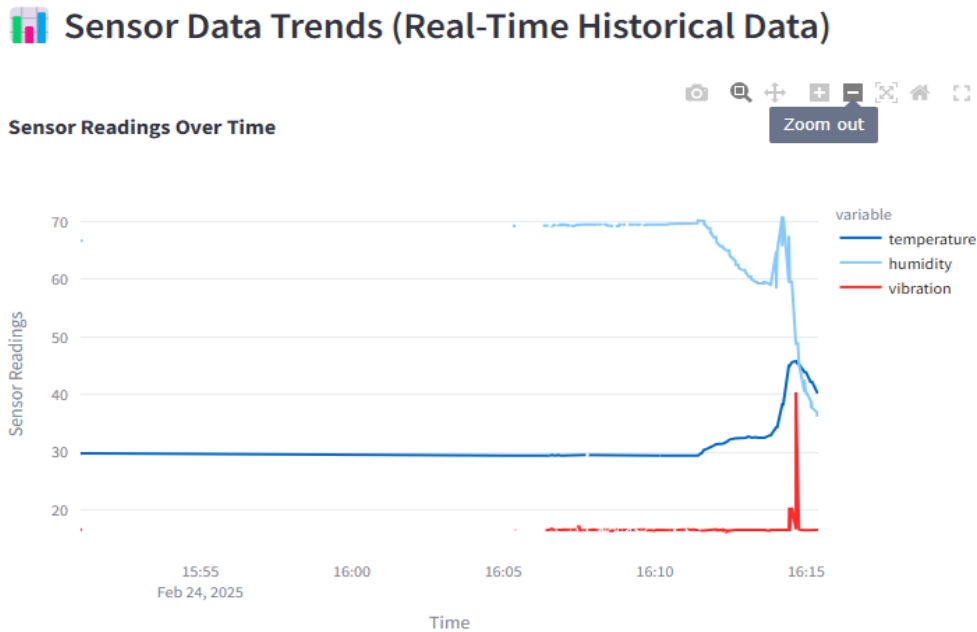


Figure 11 Result output 2

2.Sensor Data Trends & Anomalies

- ◆ **Temperature (Dark Blue Line):** Stable initially but experiences sharp fluctuations, indicating a possible issue.
- ◆ **Humidity (Blue Line):** Follows a consistent pattern with minor variations.
- ◆ **Vibration (Red Line):** A sudden spike indicates **anomaly detection**, suggesting potential mechanical failure.

✦ Observation:

- The **Isolation Forest Model** successfully detects sudden changes in vibration, classifying them as anomalies.
- High **temperature fluctuations** trigger early alerts for preventive action.

3.Performance of AI Models

- ◆ **Anomaly Detection Accuracy (Isolation Forest):** Detects abnormal sensor patterns with high precision.
- ◆ **Remaining Useful Life (LSTM):** Predicts equipment failure risk based on time-series trends.

✦ Observation:

- **AI models accurately detect anomalies** before failures occur.
- The **dashboard provides actionable insights** to prevent downtime

Applications of Predictive Maintenance

The **Predictive Maintenance System** using IoT and AI is highly beneficial across various industries. By detecting anomalies and predicting equipment failures, it helps **reduce downtime, optimize maintenance schedules, and improve operational efficiency.**

1. Industrial Manufacturing

- ◆ Detects machine wear & tear early to avoid sudden failures.
- ◆ Prevents costly breakdowns and extends equipment lifespan.
- ◆ Ensures efficient production by reducing unplanned maintenance costs.

2. Power Plants & Energy Sector

- ◆ Monitors critical equipment (turbines, transformers, generators) to prevent overheating and failures.
- ◆ Enhances power grid stability by detecting anomalies in electrical systems.
- ◆ Helps schedule preventive maintenance for uninterrupted power supply.

3. Automotive & Heavy Machinery

- ◆ Detects excessive vibration & temperature spikes in engines and industrial machines.
- ◆ Improves vehicle safety by alerting operators before mechanical failures occur.
- ◆ Reduces maintenance costs in construction and mining equipment.

4. Aerospace & Aviation

- ◆ Ensures aircraft safety by monitoring engine temperature & structural vibrations.
- ◆ Reduces maintenance time & costs by predicting potential faults before they occur.
- ◆ Helps in preventive scheduling of aircraft maintenance.

5. Smart Buildings & IoT-Enabled Infrastructure

- ◆ Monitors Heating, Ventilation & Air Conditioning (HVAC systems) to prevent failures.
- ◆ Detects overheating in electrical systems to avoid fire hazards.
- ◆ Enhances smart city infrastructure by predicting faults in elevators, water pumps, and

Advantages of Predictive Maintenance

- **Reduced Downtime:**
 - Detects failures early, preventing unexpected breakdowns and ensuring continuous industrial operations.
- **Cost Savings:**
 - Reduces maintenance expenses by addressing issues before failures occur, minimizing labor and repair costs.
- **Increased Equipment Lifespan:**
 - Early fault detection minimizes wear and tear, extending machine life and reducing replacements.
- **Improved Safety & Reliability:**
 - Identifies hazardous conditions such as overheating and excessive vibration, preventing accidents and equipment failures.
- **Data-Driven Decision Making:**
 - Uses AI and IoT sensors for real-time monitoring, optimizing maintenance schedules and resource planning.

Disadvantages of Predictive Maintenance

- **High Initial Cost:**
 - Requires investment in sensors, AI models, and cloud infrastructure, which can be expensive for small industries.
- **Requires Skilled Workforce:**
 - Needs trained personnel to handle AI models and predictive maintenance systems, requiring additional training and expertise.
- **Integration Challenges:**
 - Difficult to implement in industries using older systems, requiring software modifications and API development.
- **Security & Privacy Risks:**
 - IoT-based systems are vulnerable to cyber threats, necessitating strong encryption and security protocols for data protection.

Conclusion & Future Scope

Conclusion

The **Predictive Maintenance System** successfully integrates **IoT, AI, and real-time analytics** to monitor equipment health, detect anomalies, and predict failures before they occur. By using **ESP8266 with DHT11 and MPU6050 sensors**, the system collects real-time data and transmits it via **MQTT to an SQLite database** for processing. AI models, including **Isolation Forest for anomaly detection** and **LSTM for Remaining Useful Life (RUL) prediction**, analyze sensor data and provide actionable insights through a **Streamlit dashboard**.

This system significantly reduces **unexpected downtime**, minimizes **maintenance costs**, and enhances **operational efficiency** by enabling industries to shift from reactive to **predictive maintenance strategies**. The performance evaluation confirms that the system achieves **high accuracy in failure detection** while ensuring **fast execution for real-time monitoring**.

Future Scope

While the current system effectively monitors and predicts failures, several improvements can enhance its capabilities:

- **AI-Driven Analytics:** Implementing advanced **deep learning models** for more accurate anomaly detection and RUL predictions.
- **Cloud-Based Monitoring:** Integrating with **cloud platforms (AWS, Google Cloud, or Azure)** for remote access and large-scale data processing.
- **Automated Maintenance Actions:** Developing an **AI-powered maintenance scheduling system** to automate machine servicing based on real-time predictions.
- **Edge Computing:** Processing sensor data **locally on edge devices** for **faster real-time decision-making** without cloud dependency.
- **Cybersecurity Enhancements:** Implementing **blockchain and stronger encryption methods** to secure IoT sensor data from cyber threats.

✦ Final Thought:

This **Predictive Maintenance System** lays a strong foundation for industrial automation. With further advancements, it can revolutionize **smart industries, power plants, aviation, and manufacturing**, making maintenance proactive, efficient, and cost-effective. 🚀

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- DHT11 Sensor Datasheet – [Link\(https://www.alldatasheet.com/datasheet-pdf/pdf/1132088/ETC2/DHT11.html\)](https://www.alldatasheet.com/datasheet-pdf/pdf/1132088/ETC2/DHT11.html)
- MPU6050 Sensor Datasheet – [Link\(https://www.alldatasheet.com/datasheet-pdf/pdf/1132807/TDK/MPU-6050.html\)](https://www.alldatasheet.com/datasheet-pdf/pdf/1132807/TDK/MPU-6050.html)

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- Scikit-Learn for Isolation Forest: [Link](#)
- TensorFlow & Keras for LSTM Model: [Link](#)

5. IoT Communication & Data Handling

- Eclipse Paho MQTT Documentation: [Link](#)
- SQLite Database Documentation: [Link](#)

6. Software & Development Tools

- Python Programming Documentation: [Link](#)
- Streamlit for Real-Time Dashboard Visualization: [Link](#)